

Sameeul Khondoker Jisan

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Research Profile

Data Science-focused Software Engineering student (CGPA **3.55/4.00**), majoring in Data Science at Daffodil International University. Active researcher with two in-progress publications in machine learning and predictive analytics; experienced in end-to-end ML workflows encompassing data preprocessing, statistical feature engineering, model evaluation, and SHAP-based Explainable AI (XAI). Dual Data Science internship experience; IBM-certified in **Data Analytics** and **Data Science**. Seeking **research assistant** positions, **graduate study** opportunities, and advanced internships in **Data Science**, **Analytics**, **AI**, and **Machine Learning**.

Research Interests

- Machine Learning & Predictive Modelling
- Explainable AI (XAI) & Model Interpretability
- Data Quality & Missing Data in ML Systems
- Environmental AI & Sustainability Analytics
- Natural Language Processing
- AI-Powered Intelligent Systems
- Business Analytics
- Data Visualisation & Dashboarding
- Business Intelligence

Education

B.Sc. in Software Engineering — Major: Data Science **JAN 2023 – Present**
[Daffodil International University](#)

- CGPA: **3.55/4.00** | Relevant Coursework: Machine Learning, Data Mining, Statistics, Database Systems, Algorithm Design, Software Engineering

Higher Secondary School Certificate (HSC) — Science **JUL 2019 – DEC 2021**
[Daffodil International College](#)

- GPA: **4.33/5.00**

Secondary School Certificate (SSC) — Science **JAN 2017 – FEB 2019**
[Agargaon Adarsha High School](#)

- GPA: **3.89/5.00**

Research & Publications *(In Progress)*

[AI-Based Predictive Modeling of Individual Carbon Footprints Using Behavioral and Energy Consumption Data](#)

- Designed and evaluated ensemble ML models (Random Forest, XGBoost) and a neural network regressor (MLP) to predict individual carbon footprints from behavioural and energy-consumption datasets; conducted statistical performance benchmarking across model configurations.
- Applied systematic data preprocessing, feature engineering, and **SHAP-based Explainable AI (XAI)** analysis to identify the most influential environmental and lifestyle predictors; evaluated model outputs using cross-validation and standard regression metrics.

[When Machine Learning Models Should Not Predict: A Study on the Effect of Incomplete Data on Machine Learning Performance](#)

- Investigated the quantitative impact of data incompleteness on supervised learning model accuracy and reliability through controlled experimental design with systematically varied missingness rates and data-quality conditions.
- Benchmarked Logistic Regression, Decision Tree, and K-Nearest Neighbors classifiers across varying incompleteness levels; documented accuracy degradation curves and proposed **data-quality risk thresholds** for production ML deployment decisions.

Technical Projects

Finzy — AI-Powered Personal Finance Management System

github.com/mrkhondoker/finzyy_ai-assisted_expense_management_system

Stack: Next.js, PostgreSQL, Supabase, Google Gemini AI, Docker, REST API

- Engineered an AI-powered financial management platform integrating **Google Gemini AI** for automated receipt OCR and intelligent expense categorisation; implemented a real-time financial analytics dashboard with budget tracking and **AI-driven spending insight** generation.

Heart Failure Prediction

github.com/mrkhondoker/heart-failure-prediction

Stack: Python, Scikit-learn, Pandas, Matplotlib, Seaborn

- Applied Logistic Regression to **12 clinical biomarkers** from a real-world clinical dataset, achieving **84% classification accuracy**; conducted comprehensive **EDA**, correlation analysis, and model evaluation using confusion matrix, precision-recall, and F1-score reporting.

Student Performance Prediction

github.com/mrkhondoker/student-performance-prediction

Stack: CatBoost, Scikit-learn, Flask, Docker

- Developed an end-to-end **predictive analytics pipeline** using CatBoost gradient boosting to identify key academic performance drivers; model served via **Flask REST API** with Dockerised deployment for scalable inference.

Movie Recommendation System

github.com/mrkhondoker/movie-recommender-system

Stack: Flask, Scikit-learn, Pandas, NumPy, SciPy, TMDB API

- Implemented a **hybrid recommendation engine** combining content-based filtering (TF-IDF, cosine similarity) and collaborative-filtering techniques; integrated TMDB API for live data retrieval; deployed as a full-stack web application.

Work Experience

Data Science Intern | TecVix

MAY 2026 – Present

- Developing production **ML pipelines** incorporating data preprocessing, **feature selection**, and predictive modelling; applying statistical analysis to extract actionable insights from real-world datasets to support data-driven product and BI research decisions.

Data Science Intern | SaiKet Systems

APR 2026 – MAY 2026

- Processed **50K+ records** via automated Pandas/NumPy data pipelines, reducing preprocessing time by **~40%**; performed **EDA**, data cleaning, and statistical analysis to support business intelligence initiatives.
- Deployed **3 ML models** via REST API, improving BI reporting accuracy by **~30%**; applied statistical techniques and Python-based tools to develop data workflows and contribute to AI/ML projects.

Video Editor & Content Creator | GoEdu

FEB 2025 – FEB 2026

- Produced **50+ educational videos** reaching **10K+ learners**; applied **data-informed content strategy**, driving a **25%** engagement increase through performance analytics and iterative improvement.
- Collaborated with cross-functional teams on multimedia development supporting professional skill development and digital learning initiatives.

Volunteer | Career Development Center (CDC-DIU)

MAR 2025 – MAY 2026

- Coordinated logistics for job fairs and career development events; assisted in organising professional development workshops and managing participant coordination, communication, and team activities.

Data Entry Operator | Golden Harvest InfoTech Limited

JAN 2022 – DEC 2022

- Managed and verified **100K+ records** at **99.8%** accuracy; optimised data verification workflows, reducing processing time by **20%**, ensuring consistency across enterprise datasets.

Technical Skills

Data Science & Statistical Analysis: Pandas, NumPy, Statistical Analysis, EDA, Feature Engineering, Data Preprocessing, Predictive Modelling, Data Cleaning

Data Visualisation & BI: Matplotlib, Seaborn, Plotly, Tableau, Power BI

Programming & Query Languages: Python, SQL, R, Java, C/C++, JavaScript

Machine Learning & AI: Scikit-learn, XGBoost, CatBoost, Random Forest, Logistic Regression, Decision Tree, KNN, MLP Regressor, NLP, Generative AI

Model Evaluation & Explainability: SHAP (XAI), Cross-Validation, Confusion Matrix, Classification Reports, Regression Metrics, Hyperparameter Tuning

Databases & Deployment: PostgreSQL, MySQL, Supabase, Flask, Spring Boot, Docker, REST API

Research & Development Tools: Jupyter Notebook, Google Colab, Git/GitHub, VS Code, PyCharm, Google Cloud

Awards & Competition

- **DIU Job Utsob 2024 & 2025:** Recognised as an active volunteer for outstanding contributions in event logistics, coordination, and career development programme management at Daffodil International University.
- **DIU CSE Fest 2025 Project Show:** Presented innovative software and technology-based solutions in a competitive academic showcase.
- **GoEdu Campus Ambassador Programme:** Awarded **Top Campus Ambassador** for exceptional leadership, community engagement, and student outreach contribution.
- **National Robotics Championship 2025:** Competed at national level, demonstrating teamwork, technical problem-solving, and innovation skills.

Certifications

- [IBM Data Science Professional Certificate](#) — Coursera
- [IBM Data Analyst Professional Certificate](#) — Coursera

Language Proficiency

- **IELTS Academic:** Overall Band Score **7.0** (Listening 7.5, Reading 7.0, Writing 7.0, Speaking 6.5)

Extracurricular Activities

- Campus Ambassador, **GoEdu**.
- Campus Ambassador, **Career Development Center (CDC-DIU)**.
- Contributed to transport management, DIU Job Utsob 2024; logistics coordination, DIU Job Utsob 2025.
- Organising team member, **DIU SEC Code Trap** programming competition.